THE OBSTRUCTION CLASS FOR DEFORMING A MORPHISM OF RINGED SPACES OVER A THICKENING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the situation and the equation-to-solve fixed in the preceding section; the proof recalls the two identifications it uses.

Editorial note. The eight references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The identification of the naive cotangent complex with the truncation of the cotangent complex for ringed spaces. Lemma lemma-compare-cotangent-complex-with-naive-ringed-spaces, used at proof lines 3854-3856. Transfers the deformation-theoretic statements.
- The identification H^0(L_{X/S}) = Omega_{X/S}. Lemma lemma-H0-L-morphism-ringed-spaces, used at proof lines 3857-3859. Is the second identification used for parts (2) and (3).
- The deformation-theory statements for morphisms of ringed spaces. Deformation Theory, Lemmas defos-lemma-parametrize-solutions-ringed-spaces, defos-lemma-huge-diagram-ringed-spaces and defos-lemma-choices-ringed-spaces, used at proof lines 3859-3861 and 3864-3870. Supply parts (2), (3) and the class xi'.
- The distinguished triangle for X -> S -> S'. Lemma lemma-triangle-ringed-spaces, used at proof lines 3884-3891. Gives the long exact sequence producing the obstruction.

Source reference. Original lemma 08UC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11195–11195. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08U8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11191–11191. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0GQ3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20848–20848. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original remark 0GQ4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20849–20849. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08UY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11216–11216. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08UV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11213–11213. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08UW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11214–11214. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08T4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11152–11152. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *In the situation above we have*

(1) *There is a canonical element* $\xi \in \mathrm{Ext}^2_{\mathcal{O}_X}(L_{X/S}, \mathcal{G})$ *whose vanishing is a sufficient and necessary condition for the existence of a solution to (08UY).*
(2) *If there exists a solution, then the set of isomorphism classes of solutions is principal homogeneous under* $\mathrm{Ext}^1_{\mathcal{O}_X}(L_{X/S}, \mathcal{G})$.
(3) *Given a solution* X'*, the set of automorphisms of* X' *fitting into (08UY) is canonically isomorphic to* $\mathrm{Ext}^0_{\mathcal{O}_X}(L_{X/S}, \mathcal{G})$.

Proof. Via the identifications $NL_{X/S} = \tau_{\geq -1} L_{X/S}$ (Lemma 08UW) and $H^0(L_{X/S}) = \Omega_{X/S}$ (Lemma 08UV) we have seen parts (2) and (3) in Deformation Theory, Lemmas 08U8 and 08UC.

Proof of (1). Roughly speaking, this follows from the discussion in Deformation Theory, Remark 0GQ4 by replacing the naive cotangent complex by the full cotangent complex. Here is a more detailed explanation. By Deformation Theory, Lemma 0GQ3 there exists an element

$$\xi' \in \mathrm{Ext}^1_{\mathcal{O}_X}(Lf^* NL_{S/S'}, \mathcal{G}) = \mathrm{Ext}^1_{\mathcal{O}_X}(Lf^* L_{S/S'}, \mathcal{G})$$

such that a solution exists if and only if this element is in the image of the map

$$\mathrm{Ext}^1_{\mathcal{O}_X}(NL_{X/S'}, \mathcal{G}) = \mathrm{Ext}^1_{\mathcal{O}_X}(L_{X/S'}, \mathcal{G}) \longrightarrow \mathrm{Ext}^1_{\mathcal{O}_X}(Lf^* L_{S/S'}, \mathcal{G})$$

The distinguished triangle of Lemma 08T4 for $X \to S \to S'$ gives rise to a long exact sequence

$$\ldots \to \mathrm{Ext}^1_{\mathcal{O}_X}(L_{X/S'}, \mathcal{G}) \to \mathrm{Ext}^1_{\mathcal{O}_X}(Lf^* L_{S/S'}, \mathcal{G}) \to \mathrm{Ext}^2_{\mathcal{O}_X}(L_{X/S}, \mathcal{G}) \to \ldots$$

Hence taking ξ the image of ξ' works. □